Documenting Electronics in DocDB

The following slides show how to:

- Create a numbering scheme within DocDB to handle initial release, and revisions of electronics

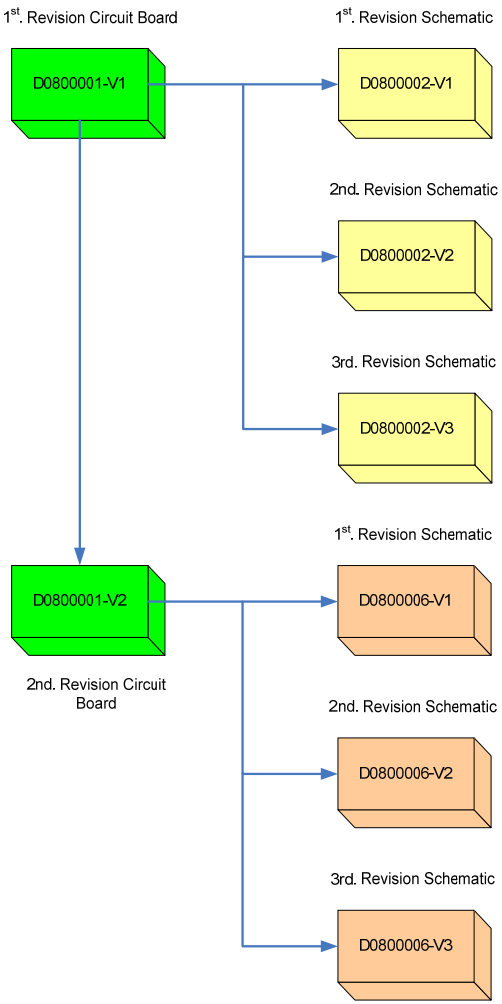
- Put sufficient data on circuit boards and schematics to be able to cross-reference physical circuit boards to documentation and vice-versa

- Understand the minimum data that must be submitted to properly document a design

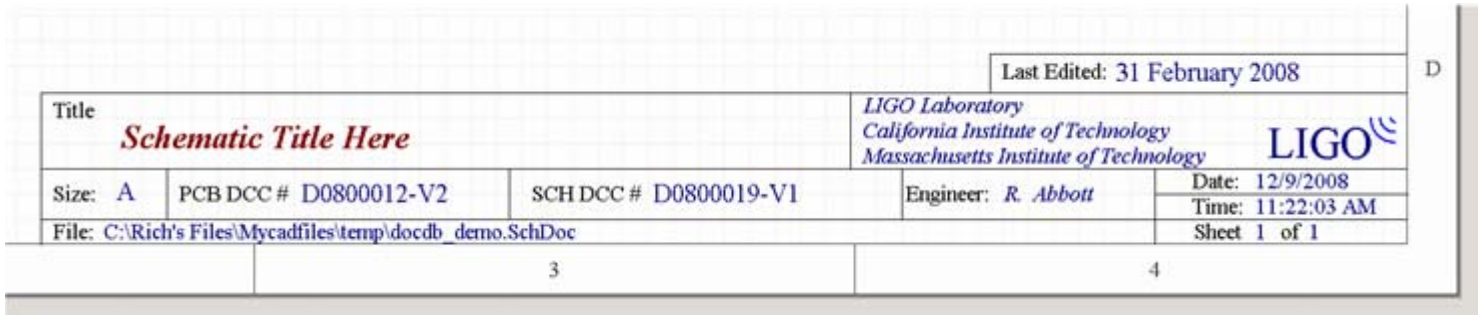
- Understand the new Serial Number system for circuit boards and chassis

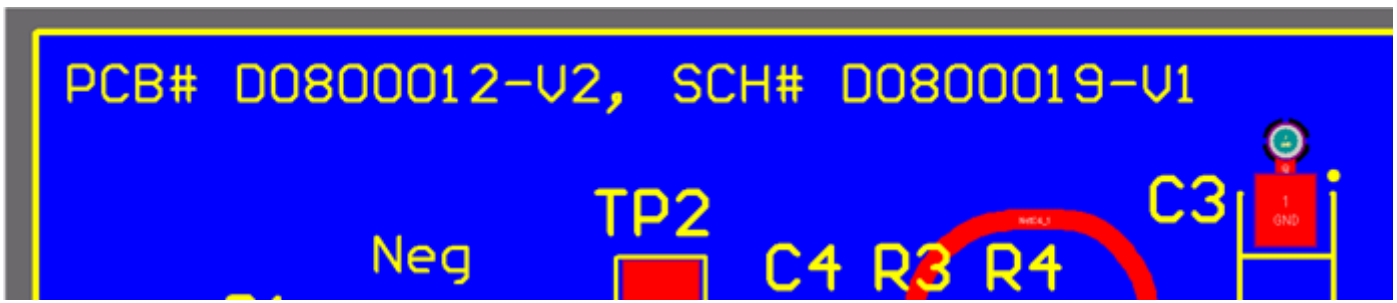
- Use the E-traveler paperless traveler form.

PCB and Schematic Revision Flow



What to Put on Schematics and Circuit Boards





Minimum Data for Documentation

To properly document a design, a couple of fundamental items must be archived:

- PDF version of the schematics and board layout
- A zipped file containing the native Altium Project files used in the design

To use a performance appraisal related moniker, this barely gets you to the level of “Meets Expectations”. To improve on the bare minimum, include the following:

- E-traveler
- Digital photographs of the design
- Bill of materials
- Test results
- Quick-start guide
- Gerber data used to manufacture the circuit board
- Etc. as dictated by good engineering practice

Serial Numbers for Electronics

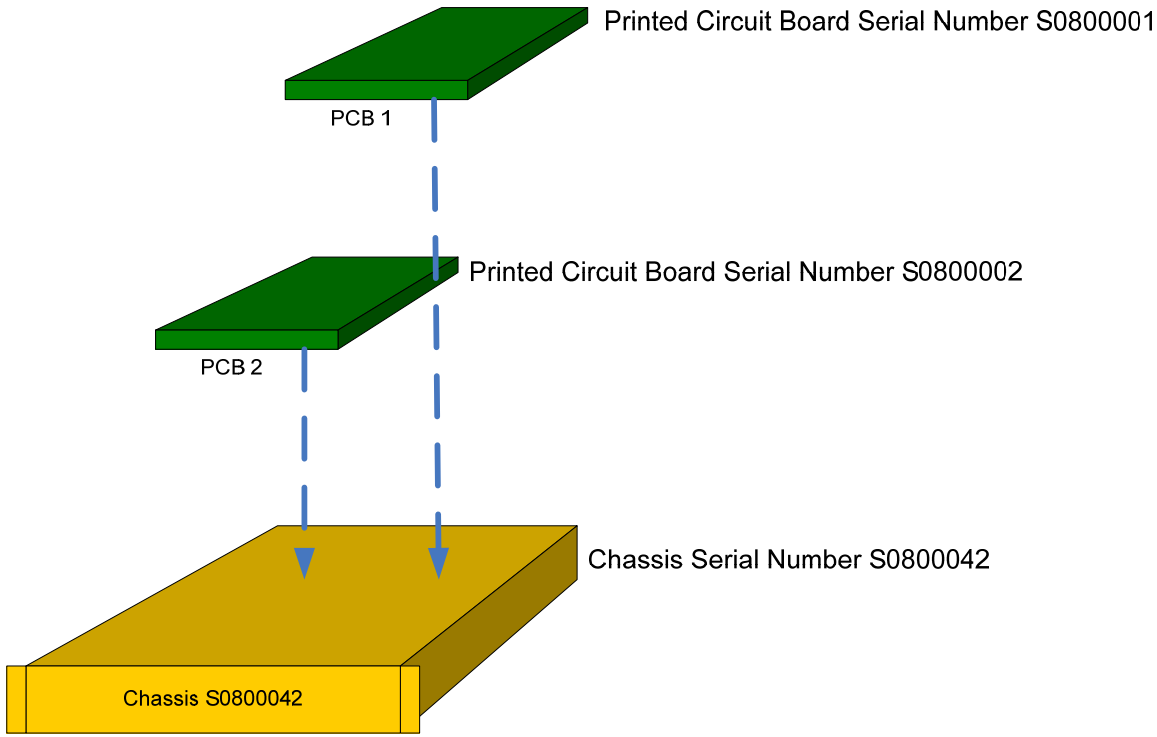
Historically, we have been making-up our own serial numbers without the benefit of a centrally coordinated source. Within DocDB, there is provision to automatically generate serial numbers that preclude the possibility of duplication.

What do these serial numbers do?

- Provide a unique identifier for circuit boards and chassis
- Link hardware to the paperless traveler system known as an E-traveler
- The serial number also acts as the document number for the E-traveler. More on this later.

Serial Numbers for Electronics, con't.

Here's an example of serial numbers in use



Anatomy of a Serial Number

Here's what makes the serial numbers tick, consider S0800005-V1 and S0900012-X0:

- The first letter "S" designates the strings as serial numbers
- The next two digits indicate the year, 2008 or 2009 in the examples
- The following five digits are the meat of the unique serial number. These range from 0 to 99,000 for a given year. Hope we don't run out.
- The last two digits indicate version number in the case of a "V", and a number that has not yet had data uploaded to the DCC in the case of X0
- As serial numbers are by nature unique, versioning doesn't make sense. The version field becomes critically useful in implementing the E-traveler

Electronic Travelers

An Electronic Traveler, or E-traveler, serves as the birth-certificate and life history for a piece of electronics.

For now, a copy can be obtained from a link on:
http://www.ligo.caltech.edu/~abbott/pages/docs.htm
The form uses a PDF based form created in Adobe Designer.

Once the form is filled out, it is saved as a PDF with the same serial number, or S-number, as the device it is documenting. For example, chassis serial number S0800005 requires a traveler. The E-traveler form is filled out and saved to file name S0800005-V1, then the form is submitted to the DCC for archive.

If this same chassis is modified in the field, the associated E-traveler is obtained from the DCC, modified, then re-submitted as revision V2, and so on through the life of the chassis.

An E-traveler can be attached to any item that has a serial number.

LIGO E-TRAVELER

Beware, the reset button removes all user entered text Reset Form

"S" Number & Revision , use for serial number and DCC number

Title of device (i.e. "Anti-Aliasing Chassis" or "Low Noise Amplifier Board" etc.)

E-Traveler Originator & Institution

Date of most recent update

Basic Instructions

This form documents the life history of a chassis, circuit board, or other assembly. The intent is to capture original manufacturing notes as well as incremental changes made to the assembly over its operating life. The text boxes have essentially unlimited depth. A scroll-bar will appear when needed. Once a change has been entered, this form should be saved as a new revision and re-submitted to the DCC. For example, E-Traveler number S080123-03 was updated and became S080123-04.

Initial Manufacturing Notes

Details of Modifications made during initial manufacture. Note all Printed Circuit Board and schematic revisions used in this design.

Performed By:

Date

Acceptance Testing

Test Procedure DCC Number & Revision

Performed By:

Date

List any discrepancies or deviations from limits established in the test procedure

What does an E-traveler Contain?

Here's a quick summary of the information that can be viewed in an E-traveler:

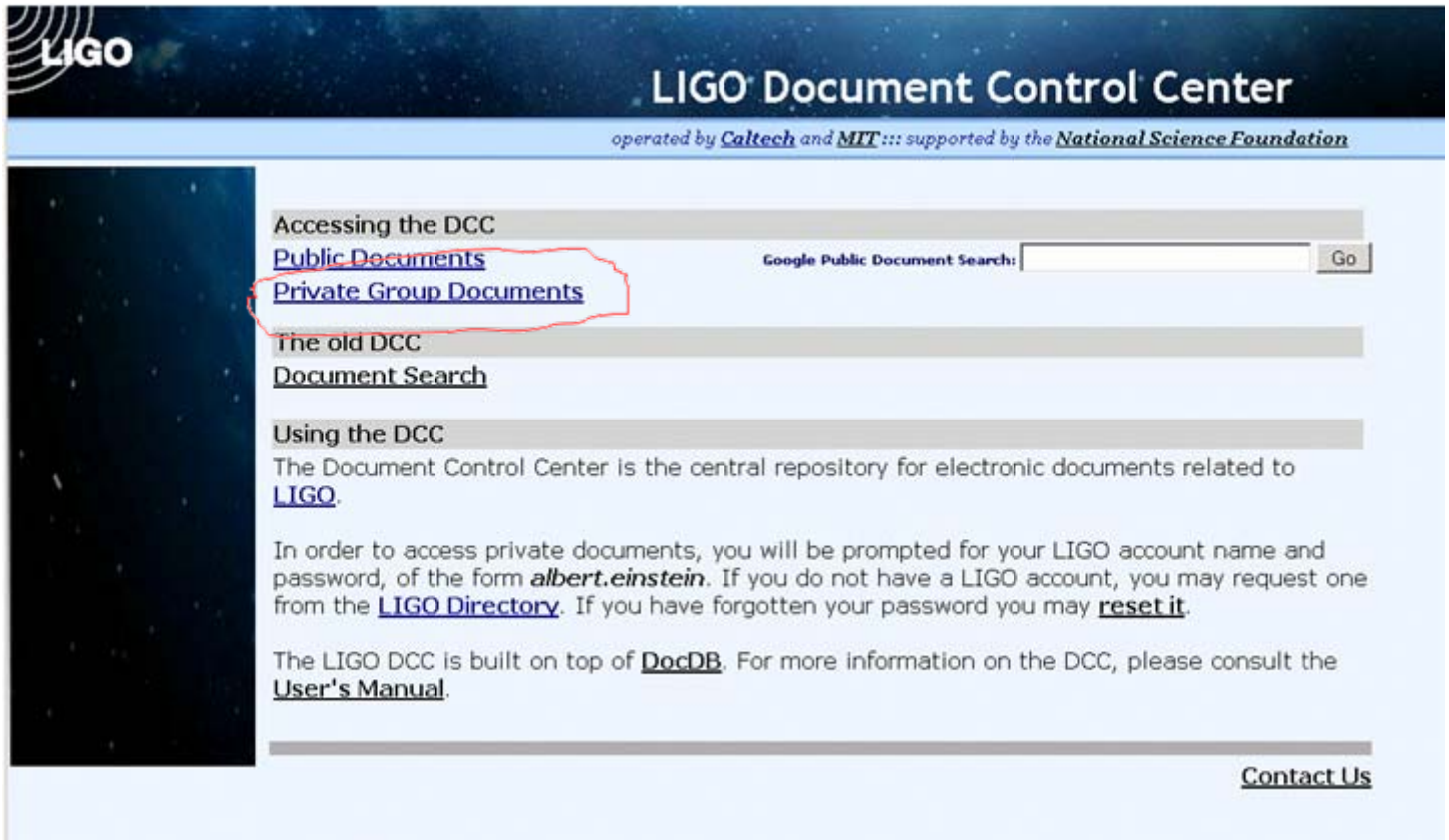
- Most recent date of change
- Revision identification of printed circuit boards and schematics
- Initial manufacturing notes
- Initial testing summary
- Physical (geographic) location of the item
- Record of all changes made to the item over its lifespan

The form is not very rigid (yet), so there is flexibility, be that good or bad, in the data fields.

Pedestrian's Guide to DocDB

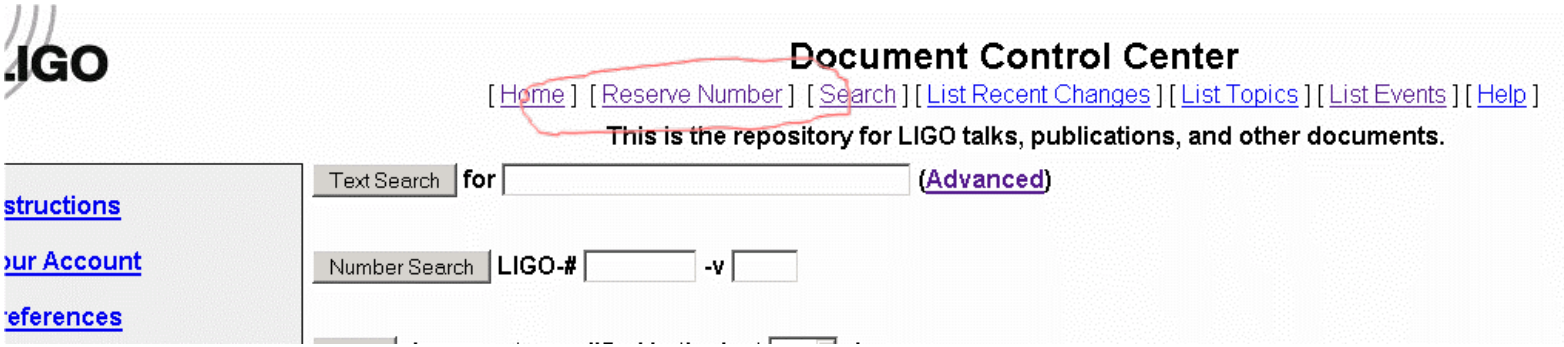
The new DCC can be found at: https://dcc.ligo.org/ It requires the login name and password associated with ligo.org

Here's what you see when you go there, click on the stuff circled in RED:



Pedestrian's Guide to DocDB (con't)

You should now see the screen below. Click on "Reserve Number" circled in RED.



The rest is easy, just follow the directions and soon you will have a document number. Once you have a number, you can either immediately upload, or come back later to upload.

Fields are available to link your document to other documents that are already in the DCC. These links can be modified, created or updated at any time now or later.